

User as Engram: Internalizing Per-User Memory as Local Parametric Edits

Anonymous submission

Abstract

Personal memory requires both mutable user facts and reusable reasoning. Retrieval-augmented generation (RAG) keeps facts external but pays a search or tool hop and serializes retrieved text at query time; per-user LoRA removes that path but stores each user as a global model update. We propose **User as Engram**: write content into hash-addressed memory rows and learn reasoning once in a shared adapter. Against per-user LoRA, the design provides three benefits. At 100 facts it uses 88 KB rather than 14.2 MB ($161\times$ less state); closed-form writes and addressed deletion avoid per-user retraining; and it matches direct recall while improving mean indirect reasoning by $5.6\times$, with its largest LOCOMO gains on multi-hop questions. Writes are inspectable and cause about $33,000\times$ less held-out loss increase. Against RAG, Engram offers an in-network lookup opportunity at every prefill and decode position, requiring neither external retrieval nor serialized memory tokens. On our controlled split, same-base RAG top-3 performs one search, adds about 29 memory tokens, and reaches 39.0% indirect accuracy versus Engram’s 44.5%. A $2.5\times$ larger Qwen-RAG top-3 leads at 34 facts, ties at 100, and falls to 30.0% versus 44.5% at 1,000 facts as retrieval recall degrades. Engram thus targets compact, frequently accessed user state, while RAG remains complementary for provenance and open-domain evidence.

Introduction

Personalization requires both remembering a user’s facts and using them to reason. Retrieval-augmented generation (RAG) keeps those facts editable, but recall takes an external systems path: retrieve from a store, serialize the result as text, and pay the resulting context and prefill cost (Lewis et al. 2020; Gao et al. 2023). In an agentic implementation, retrieval also entails generating and executing a tool call. Repeated memory access thus means repeated search or tool round trips. A per-user LoRA (Hu et al. 2022) removes that external path, but turns records into a model-shaped global update: it requires per-user optimization, provides no address for updating or deleting one fact, consumes 14.2 MB at rank 64 even for a small record set, and can perturb unrelated behavior.

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We propose *User as Engram*, a middle ground between external retrieval and undifferentiated model weights. We store user content as local edits to the hash-addressed memory table of an Engram-pretrained Transformer (Cheng et al. 2026), while learning reasoning skill once in a shared LoRA. At every token position in prefill or decoding, suffix N -grams can address memory rows whose gated values enter the residual stream in the same forward pass. Memory can therefore supply an in-network signal throughout generation, without semantic search, a tool round trip, or retrieved context tokens. This is a higher-bandwidth interface to frequently accessed user state than passing a few retrieved text snippets through the prompt; it complements rather than replaces RAG’s strengths in provenance and open-domain evidence.

Three properties make this interface attractive as a new form of user memory. *Storage*: a 100-fact override map is 88 KB, $161\times$ smaller than a rank-64 per-user LoRA. *Lifecycle*: the representation does not intrinsically require training a per-user adapter—closed-form insertion costs <1 ms, and deletion removes addressed rows with no gradient steps; optional row-only refinement can improve recall when facts interact. *Use*: local facts remain available as inputs to one transferable reasoning skill: across three seeds, Engram plus shared skill reaches 41.2% indirect accuracy versus 7.3% for per-user LoRA, and Engram’s largest LOCOMO gains occur on multi-hop questions. Figure 4 summarizes these three benefits. A fact activates only the rows selected by its suffixes; all other rows and backbone weights remain bit-identical. Small override maps can be swapped per request or combined additively, separating rapidly changing, private content from slowly learned, population-level skill, as in complementary biological memory systems (Semon 1921; Tonegawa et al. 2015; McClelland, McNaughton, and O’Reilly 1995).

Figure 1 illustrates this separation. A user’s doctor or preference may change and require individual deletion; inferring that a cardiologist is relevant to a heart check-up is stable and transferable. A per-user adapter conflates these lifecycles, repeatedly relearning the relation and making individual records difficult to locate or delete. Our design keeps content explicit and reasoning reusable.

Contributions. We make three contributions. First, we formulate per-user memory as explicit overrides to an Engram model’s hash-addressed rows, with closed-form in-

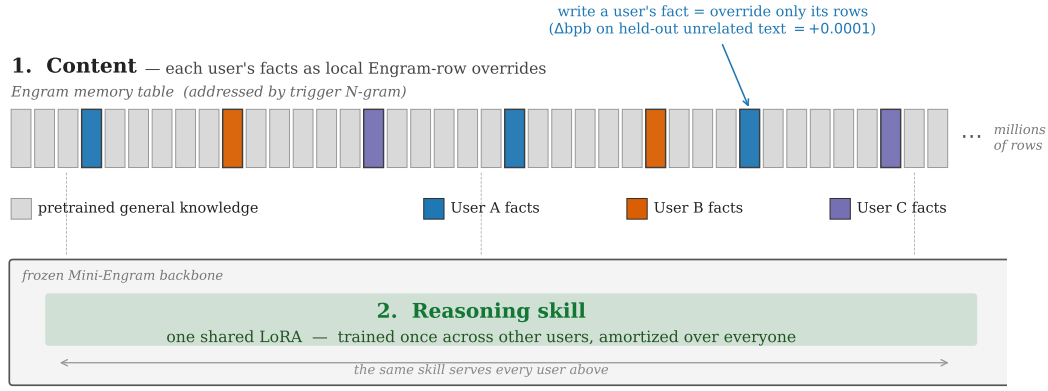


Figure 1: Local Engram content with a shared Transformer and reasoning adapter.

sertion, addressed deletion, and composable maps. Second, we separate this mutable content from one shared reasoning adapter and verify the interface mechanistically and behaviorally, including indirect and multi-hop use. Third, we provide controlled comparisons with both per-user LoRA and RAG across state, update cost, locality, reasoning, retrieved-memory tokens, and knowledge-base scaling: Engram uses $161\times$ less per-user state than LoRA and no external memory hop, while the $2.5\times$ larger Qwen-RAG baseline falls below it beyond 100 facts (Figures 4, 7, and 8).

Background and Design Requirements

Personal-memory mechanisms span three families. Prompt memories and retrieval stores keep content editable, but consume context and select evidence at every turn (Lewis et al. 2020; Gao et al. 2023). Parameter-efficient tuning moves content into a shared function (Hu et al. 2022). Model editing targets particular associations, yet usually produces weight changes whose locality is empirical (Meng et al. 2022, 2023). Engram pretraining supplies a fourth substrate: explicit key-value rows in the residual stream (Cheng et al. 2026).

We evaluate personal-memory methods along six requirements that are easy to conflate. *Recall* asks whether the exact fact can be produced. *Use* asks whether it can participate in a novel inference. *Locality* asks whether the write leaves unrelated behavior unchanged. *Lifecycle* requires individual update and deletion. *Serving* requires many users to share one backbone. *Access* asks whether each recall needs search, a tool round trip, or extra context. A method that succeeds on recall alone can fail all five remaining requirements; this motivates reporting held-out loss, indirect questions, address-level audits, per-user storage, and query cost alongside accuracy.

The term *Engram* here denotes the model’s pretrained hashed memory table, not an arbitrary cache added after training. Its gates have learned when table values should enter the residual stream. We exploit that learned interface but do not alter its addressing rule or backbone. Our contribution is a write and composition protocol for per-user overrides, plus the separation of those overrides from a shared adapter that learns population-level reasoning.

Why Facts Should Not Be Per-User LoRA

Weights are appropriate for shared abstractions but problematic for mutable, tenant-specific records. We isolate this distinction using Mini-Engram-d20 and the same 20 users. On the canonical seed, a rank-64 LoRA trained on one fact per user raises held-out loss by 1.784 and worsens 17/20 users, whereas addressed Engram rows change loss by only 0.00005 and worsen none. Across three seeds the means are +1.698 versus +0.000052 ($\sim 33,000\times$), and 49/60 versus 0/60 users worsen.

The distinction is architectural, not merely an optimizer failure. A LoRA must move shared projections to raise the desired answer probability, so its update can fire on unrelated inputs. A keyed memory write is inert unless its address is read. Strong instruction-tuned bases can conceal the behavioral damage by already solving a question, but cannot remove the underlying loss increase.

Across Qwen2.5-3B/7B (Qwen Team 2025), Llama-3.1-8B (Grattafiori et al. 2024), and Mistral-7B (Jiang et al. 2023), only 0–20% of users score worse after personalization, and mean indirect accuracy changes by +0.088 to +0.217; on the base LM, 85% score worse and the mean is -0.133 . Instruction tuning can therefore mask the task-level effect of a global LoRA update, but the update remains nonlocal. Each per-user LoRA is still an alternative global model state, whereas sparse override maps share a backbone.

User as Engram

An Engram layer hashes token-suffix N -grams to memory rows, gates their values, and adds them to the residual stream. To write a fact, we put its answer where its trigger is read: deterministic hashes map the trigger’s suffix to a sparse address set R_f (about 16 rows for $N = 3$, $K = 8$). The backbone and every row outside R_f remain fixed. We evaluate three ways to choose the values written to R_f . A closed-form pseudo-inverse write (*UNEMBED_P*) uses the Engram value projection’s pseudo-inverse to align the row contribution with the answer token’s unembedding direction; it costs <1 ms and requires no gradient steps. Per-fact gradient optimization (*OPT*) starts from that value and minimizes answer loss by updating only R_f for each arriving fact. Joint optimization (*Joint OPT*) instead samples across a user’s facts

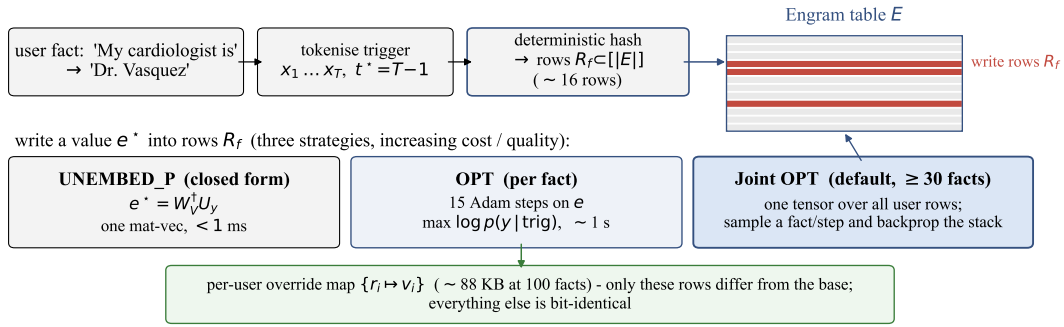


Figure 2: Surgical insertion into deterministically addressed Engram rows.

while fitting the union of their addressed rows, so earlier and co-active facts remain compatible. All three strategies have the same editable parameter set; they differ only in write compute and recall. A 100-fact override map occupies about 88 KB. Figure 2 summarizes the write process.

The edit is a glass box. Figure 3 contrasts its footprint with the matched LoRA. For an inserted fact we can enumerate the trigger hashes, verify that only R_f changed, inspect the learned gates, measure the residual contribution at each layer, and observe the answer token rise in the logit lens. In the matched diagnostic, Engram’s residual delta is exactly zero at every non-trigger position and every pre-Engram layer; the LoRA delta is nonzero at every position and layer and moves unrelated text by a mean residual norm of 107. Locality therefore follows from the data structure rather than from a regularizer.

Keys, correctness, and deletion. Personal questions often reproduce an entity or relation phrase even when surrounding wording changes. Multiple suffix orders and hashes provide redundant triggers; inserting paraphrased question forms expands coverage. This is not semantic search: a query with no overlapping trigger may miss. Because addresses are deterministic, we know the complete editable row set before optimization. Deleting a user or fact removes its override entries rather than reversing a gradient update; updating a fact replaces explicit entries. Closed-form writes need no training, while optional joint re-optimization can repair interference when facts share rows.

Mechanistic checks. We validate locality four ways: byte comparison leaves the backbone and unaddressed rows identical; non-trigger positions have zero residual delta; the trigger gate rises from 0.02 to 0.99; and the residual change aligns with the value path at cosine 0.999. The same write placed early drops recall from near-perfect to about one quarter, confirming that late placement matters.

The comparison LoRA is optimized to the same recall target. Its residual delta appears at all positions and layers, including an unrelated “capital of France” control, because low-rank does not mean locally activated. Norm or KL regularization might reduce that footprint, but would not make the update address-conditioned or identify exactly which inputs can activate it. For Engram, the zero-delta region follows directly from the lookup key.

Each user owns an override map $\Delta_u : h \mapsto v$. At inference, lookups first consult that map and otherwise fall back to the shared table. Maps with disjoint keys commute, so personal, team, and organizational memories can be stacked without merging dense weight deltas. Hash collisions are explicit and can be detected. In the current implementation the last write wins; maps whose triggers overlap should therefore remain separate.

Experiments

No pretrained Mini-Engram checkpoints are publicly available. We therefore integrate Engram lookup layers into Karpathy’s nanochat GPT-2-style backbone (Karpathy 2026; Radford et al. 2019) and pretrain all four Mini-Engram checkpoints ourselves from scratch with a nanochat-style recipe. We evaluate them against per-user LoRA, in-context facts, and retrieval, matching backbone, tokenizer, generation, and facts unless noted. Metrics cover direct and paraphrase recall, held-out LM loss, per-user storage, and indirect questions that transform stored facts.

Controlled bases vary depth and width but share the lookup. We report accuracy and held-out next-token loss because strong models can mask contamination once answers saturate. Storage excludes the backbone and shared adapter, which are amortized across tenants.

Synthetic schemas control recall and reasoning; LO-COMO adds conversational memory. Direct questions request stored answers, whereas indirect questions apply held-out relations and cannot be solved by echoing triggers. We repeat key contamination and layering results over three seeds. Writes are offline; query cost includes lookup, search, and context, and storage means serialized trainable state rather than duplicated checkpoints.

Protocol and comparison controls

All Mini-Engrams use one lookup rule and a $50K \times 256$ (51.2M-parameter) table; Table 1 lists their scales. For pre-training, we use bf16 and the nanochat budget of 12 tokens per scaling parameter. Base, XL, and XXL corpora contain 100 USER + 100 ORG facts, 1,000 of each, and 3,132 trigger templates, respectively. At 100 facts, Joint OPT uses 2,000 Adam steps at rate 0.5. The three-seed test matches rank-64 LoRA (1,500 steps, 5×10^{-4} , $\alpha = 128$) against Joint OPT (1,500 steps, rate 0.5).

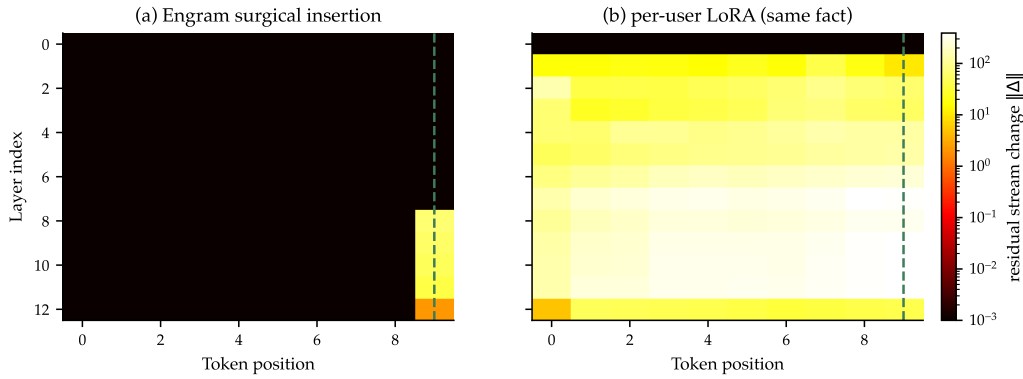


Figure 3: Residual changes from matched Engram and per-user LoRA edits.

Table 1: Mini-Engram backbones pretrained by us from scratch. All share the 51.2M-parameter memory table.

	d8 v2	d12 v2	d12@1280	d20@1536
depth $\times$ width	8×512	12×768	12×1280	20×1536
Engram layers	2, 5	2, 7	2, 7	2, 11
total parameters	178M	339M	625M	1.22B
ClimbMix tokens	0.50B	1.32B	3.34B	7.40B
validation bpb	.912	.827	.770	.730

Metrics. Direct top-1 matches the first generated token to gold; `indirect_any` finds the gold span anywhere, and LOCOMO uses token-F1. Contamination is the change in held-out bits per byte (Δ bpb) on unrelated ClimbMix: 262K tokens for the canonical run and 524,288 for three-seed runs.

The shared rank-16 adapter trains on 10 users and tests on 20 disjoint users; contamination and high-density comparisons use rank-64 per-user Q/K/V LoRA. Engram sweeps vary insertion, density, model scale, triggers, and facts; per-rank, per-seed, and capacity tables remain available for supplementary review.

No-memory measures prior knowledge. For indirect reasoning, relations are learned from other users and test content enters only through the evaluated memory, requiring unseen-content access and transferable skill. Unrelated held-out ClimbMix makes Δ bpb measure collateral change, not fact recall.

Recall, storage, and lifecycle. A single edit is within eight points of the in-context ceiling. On Mini-Engram-d12, joint optimization handles interacting addressed rows at high density, with top-5 recall above 90% to roughly 300 facts/user. At 100 facts, its sparse override occupies 88 KB versus 14.2 MB for rank-64 LoRA. Unlike a model-shaped adapter, each row remains an explicit item that can be replaced or removed; UNEMBED_P provides a no-training write path, while OPT and Joint OPT trade write compute for recall. Figure 4 summarizes these storage, lifecycle, and downstream-use distinctions.

Memory-system and LOCOMO comparisons. Nearest-neighbor retrieval is strongest when queries repeat stored triggers. Our MEM0 and MemMachine baselines use only

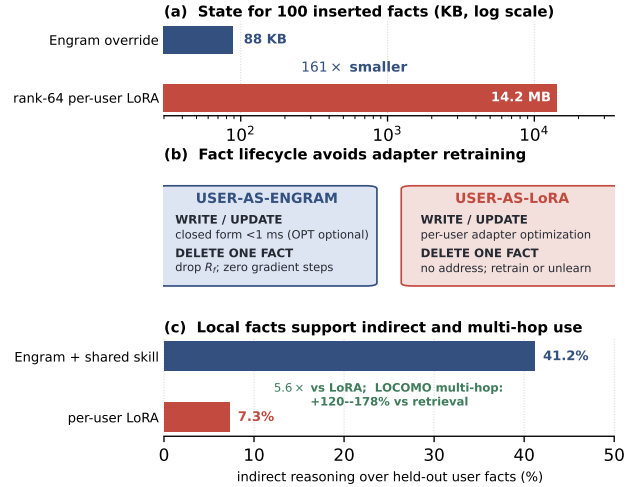


Figure 4: Engram-row benefits over per-user LoRA: (a) storage, (b) lifecycle, and (c) downstream use.

their retrieval recipes because facts are already structured; extraction and consolidation are out of scope (Chhikara et al. 2025; Wang et al. 2026). With all-MiniLM-L6-v2 (Wang et al. 2020; Reimers and Gurevych 2019), multi-trigger Engram gains 22 paraphrase points over the best retrieval baseline without context tokens.

Single-hop LOCOMO uses the first 80 non-adversarial questions from each of 10 conversations (~ 800) (Maharana et al. 2024). Retrieval stores gold evidence turns and Engram gold Q/A pairs: a controlled best-case comparison with a fixed Mini-Engram-d12 answering LM, not an end-to-end system. Across scales, Joint OPT leads on single-hop, multi-hop, and reasoning but trails on open-domain questions, where local facts cannot replace broad knowledge (Table 2; Figure 5). LongMemEval and PersonaMem-v2 provide complementary long-term and personalized memory evaluations (Wu et al. 2025; Jiang et al. 2025).

Joint OPT’s largest gains are multi-hop: token-F1 improves 120–178% over the better retrieval baseline, versus 10–58% on single-hop and reasoning. Open-domain perfor-

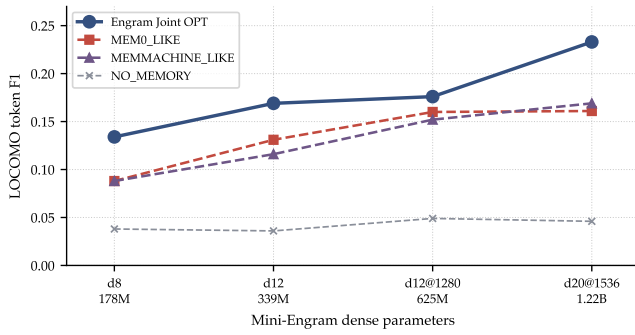


Figure 5: LOCOMO token-F1 across answering models.

mance falls 51–57%, favoring addressed personal memory plus retrieval for broad, changing knowledge.

Taken together with Figure 4’s controlled LoRA comparison, these results show that Engram facts can serve as operands for held-out transformations and multi-hop questions. Raw rows do not supply the reasoning algorithm: the key is that one shared skill can consume local facts, rather than relearning both content and skill inside every user’s adapter.

The paraphrase result separates address coverage from answer quality. Retrieval is nearly perfect on repeated prefixes; more trigger variants improve Engram paraphrases without prompt tokens. Neither implies universal semantic generalization: a stored key must activate.

Multi-hop use. Across 63 two-fact chains, fact recall is 99.2% and direct second-fact matching is 90.6%, but true chaining falls to 12.9%: the store matches but does not compose. We therefore learn relations in a shared adapter on held-out users while fitting only test-user rows. The resulting layered system reaches 41.2% indirect accuracy across seeds, $5.6\times$ the per-user-LoRA baseline; the LOCOMO category results above independently show the largest gains on multi-hop questions.

Interpreting the comparisons. Across Table 3’s three seeds, matched recall isolates storage location, density tests coexistence, and layering tests whether locality sacrifices use. Recall need not cause global contamination, and reasoning can be shared; LOCOMO shows that addressed memory does not replace open-domain retrieval.

The $33,000\times$ loss ratio is a model-, fact-, and optimization-dependent stress test, not a constant. The robust distinction is qualitative: Engram can be inactive off-address, whereas LoRA changes the shared mapping. Likewise, 88 KB is the chosen 100-fact configuration, not a lower bound; width, precision, triggers, and collisions change it.

Negative results and boundary cases. UNEMBED_P is fast but underperforms when pretrained values misalign with the answer. Independent fitting degrades under trigger collisions, and joint fitting repairs only part of the high-density loss. Free continuation is harder than short-span matching; unseen paraphrase triggers can fail. An arbitrary reasoning LoRA also fails unless it learns the same content interface,

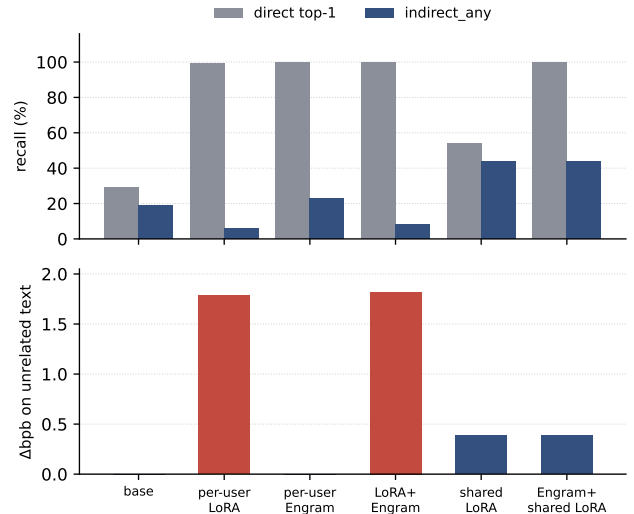


Figure 6: Direct and indirect recall and held-out Δ bpb across six conditions on Mini-Engram-d20 (seed S0, $n = 20$).

motivating layered training.

Reproducibility scope. The source retains insertion-strategy, rank, capacity, per-seed, pretraining, latency, judge-prompt, and alternative-architecture details for a planned supplement; the main paper retains the results needed for the causal chain.

Shared Skill, Local Content

We train the rank-16 shared LoRA on 10 users with each indirect question’s needed facts in context, then test 20 disjoint users whose facts arrive only through Engram rows. Six conditions cross content placement (none, per-user LoRA, or Engram) with shared skill while holding direct recall nearly constant. Across three seeds, layered Engram reaches 100% versus 98.8% direct recall and 41.2% versus 7.3% indirect accuracy for per-user LoRA.

The layered system pays $+0.385$ Δ bpb once for the shared adapter; each user’s rows add only ≈ 0.00005 , versus $+1.698$ for a separate per-user LoRA. The adapter learns a reusable transformation while local rows supply user-specific arguments (Figure 6).

The layered-minus-LoRA indirect gain has paired-bootstrap 95% CI $[+31, +37]$ percentage points.

The factorial design attributes the gain: Engram without shared skill tests recall, shared skill without content tests leakage, and naive LoRA+Engram tests arbitrary stacking. Only intended layering gives local content, transferable reasoning, and low held-out loss; train and test users are disjoint.

Cross-schema transfer. The headline comparison uses disjoint users within one schema family. To test beyond those templates, we train on first-person personal facts and evaluate on third-person medical records. Layered indirect accuracy falls from 44% to 31% (30% relative), while per-user LoRA falls from 6% to 4%; the advantage persists ($7.4\times$ to $7.6\times$).

Table 2: LOCOMO token-F1 by category and model. Joint OPT is compared with the stronger retrieval baseline in each cell.

category	answering LM	MEM0	MemMachine	Engram Joint OPT	lead vs. best retr.
single-hop	d12 v2	0.131	0.116	0.169	+29%
	d12@1280	0.160	0.152	0.176	+10%
	d20	0.161	0.169	0.233	+38%
multi-hop	d12 v2	0.092	0.076	0.243	+165%
	d12@1280	0.115	0.104	0.253	+120%
	d20	0.102	0.108	0.299	+178%
reasoning	d12 v2	0.117	0.113	0.183	+56%
	d12@1280	0.123	0.136	0.203	+49%
	d20	0.135	0.168	0.266	+58%
open-domain	d12 v2	0.247	0.209	0.122	−51%
	d12@1280	0.341	0.327	0.146	−57%
	d20	0.314	0.338	0.159	−53%

Table 3: Three-seed robustness on Mini-Engram-d20 ($n = 20$ users per seed).

Quantity	S0	S1	S2	Mean (range)
per-user LoRA Δ bp	1.784	1.754	1.557	1.698 (1.56–1.78)
per-user Engram Δ bp	.000052	.000049	.000054	.000052
layered total Δ bp	.386	.388	.379	.385 (.38–.39)
layered indirect_any	44.5%	44.2%	34.8%	41.2% (34.8–44.5)
per-user LoRA indirect_any	6.0%	8.8%	7.2%	7.3% (6.0–8.8)
worse than base (layered / LoRA)	0/20 / 17/20	0/20 / 16/20	0/20 / 16/20	0/60 / 49/60

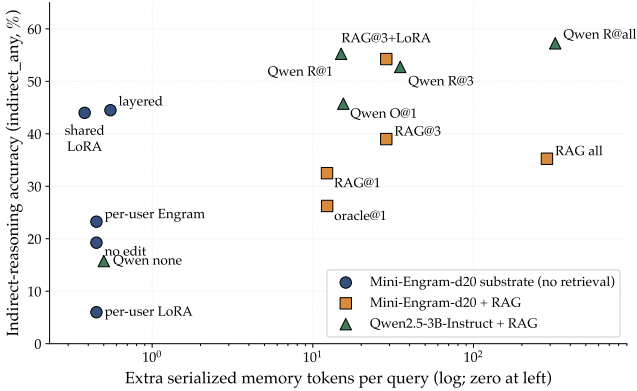


Figure 7: Engram versus RAG: indirect accuracy against extra serialized memory tokens per query; RAG points each perform one search.

Locality is unchanged: Δ bp stays +0.386 because personal rows activate only at addressed triggers.

Figure 7 compares the memory-access paths on the 34-fact split. Layered Engram reaches 44.5% indirect accuracy with no retrieved fact text or external memory call. RAG top-3 on the same answering model performs one search, serializes about 29 additional memory tokens, and reaches 39.0%; including all facts raises the memory block to 287 tokens and lowers accuracy to 35.3%. Combining RAG top-3 with the shared reasoning adapter instead reaches 54.3%, showing that retrieval and reusable skill are complementary.

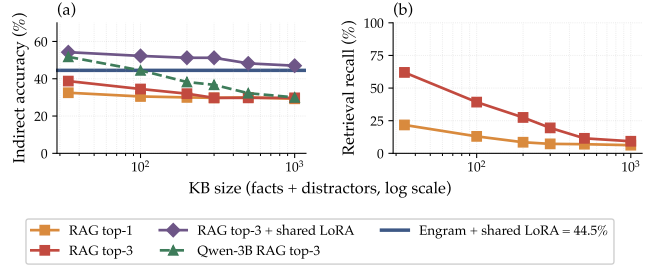


Figure 8: Engram versus RAG as the knowledge base grows: (a) indirect accuracy and (b) shared retriever recall. Qwen-3B uses a $2.5\times$ larger answering model.

The $2.5\times$ larger Qwen2.5-3B-Instruct RAG baseline (Qwen Team 2025) reaches 51.8% with top-3 retrieval and 35 added memory tokens.

The distinction is access granularity. RAG searches once and passes a small set of facts through the prompt; an agentic pipeline additionally generates and executes the tool call, and sequential recalls repeat that external hop. Engram instead computes suffix addresses at every prefill and decode position and injects selected values inside the forward pass. It therefore supplies a token-granular memory channel without retrieved context, while retaining RAG as the better complement for rapidly changing corpora, provenance, and open-domain evidence.

To isolate knowledge-base size, we add held-out schema-family distractors to each user’s 34 facts, reaching $N \in \{34, 100, 200, 300, 500, 1000\}$ with the same 20 users and

probes. The Mini-Engram RAG conditions use the same answering backbone as Engram; Qwen-RAG uses the larger instruction-tuned model. RAG top- k places the k highest-ranked facts in context, and all RAG conditions share the same retriever. Its top-3 recall falls from 62.0% at 34 facts to 9.3% at 1,000. Qwen-RAG top-3 consequently falls from 51.8% to 30.0%: it ties Engram’s constant 44.5% at 100 facts and trails thereafter. Same-base RAG top-3 falls from 38.8% to 29.8%. RAG plus the shared adapter remains slightly ahead at 1,000 facts (47.0%), again favoring a hybrid when retrieval’s provenance is needed (Figure 8).

Multi-Tenant Serving

A request resolves a sparse user table, saves and overrides affected base rows, runs one forward pass, then restores them. During that pass, each token’s suffix can fetch a gated memory value directly into the residual stream; there is no separate retrieval service or memory text to prefill. User switches change neither dense weights nor the shared adapter. Disjoint personal, team, and organizational scopes can layer; overlapping triggers are last-write-wins and should remain separate. Unmeasured batched serving would require one gather per Engram layer.

At one million users, 88 KB each is roughly 88–100 GB depending on accounting, plus one 11.8 MB shared rank-16 adapter; 14.2 MB per-user LoRAs require about 14.2 TB. Omitting allocator and index overheads, sparse content scales with stored facts while each LoRA scales with model dimensions.

S-LoRA and Punica (Sheng et al. 2024; Chen et al. 2024) batch dense adapters, but each tenant retains a model-shaped delta; overrides are indexed reads in the existing layer. Production requires untested caching, authorization, encryption, collision telemetry, and rate limits. The prototype demonstrates composition and scaling, not security.

The prototype uses sequential row swaps, not an optimized batched kernel. On one idle Blackwell GPU, d12@1280 processed 232 requests/s with 30 users and 50 facts/user (4.2 ms p50 latency), and 226 requests/s with 100 users and 100 facts/user (4.4 ms p50). These measurements are consistent with per-request work depending on the active override rather than the registered-user population, but they are not a controlled user-count scaling study because the fact count also changes. Absolute latency depends on cache residency, hardware, and the serving implementation.

Discussion, Limitations, and Related Work

At 1,000 facts Joint OPT reaches 35% top-1; with <1% slot occupancy, the ceiling is forward-pass interference among co-active rows, not collisions. Paraphrase coverage depends on triggers, and optimization is slower than appending a document.

Our work connects parameter-efficient personalization and per-user adapters (Houlsby et al. 2019; Hu et al. 2022; Zhang et al. 2024; Su et al. 2025; Tan, Liu, and Jiang 2024; Zhuang et al. 2024), retrieval-augmented generation (Lewis et al. 2020; Gao et al. 2023; Borgeaud et al. 2022), model editing (Meng et al. 2022, 2023; Mitchell et al. 2022a,b; Wang

et al. 2024), and external or semi-parametric memory (Lample et al. 2019; Packer et al. 2023; Cheng et al. 2026). These methods provide non-parametric access or global weight changes. User as Engram occupies a distinct middle point: records remain explicit and mutable like an external store, but their values are consumed inside the network without retrieval calls or context tokens. Our content/skill boundary places those mutable facts in addressed rows and reusable reasoning in shared weights, yielding locality, high-frequency memory access, and multi-tenancy rather than dense per-user models. The dense-adapter interference we measure is related to catastrophic forgetting in continual learning (McCloskey and Cohen 1989; Kirkpatrick et al. 2017).

Validity and societal considerations. Controlled models expose changed rows and small loss deltas, strengthening causal claims but limiting frontier-scale conclusions. LOCOMO mixes memory and base knowledge, so we report category reversals. Storage and latency constants are implementation-dependent, though their scaling forms are robust. Addressability aids inspection and erasure but can expose records; untested exfiltration, access control, output auditing, and shared-skill fairness remain. Generative AI tools provided editorial assistance; the authors reviewed and verified all text, claims, citations, figures, and reported results.

What should improve next. Three recipe changes target these limits: multi-fact training, random per-user hash offsets for unseen rows, and “recall, then reason” chaining examples (Sun et al. 2023). Preliminary multi-fact finetuning reached the same recall faster without raising its ceiling. Teacher traces lowered accuracy, showing that training and evaluation formats must align.

The strongest next comparison is a frontier-scale model pretrained with this interface. Small bases expose locality and failures but leave larger-scale utility untested. We expect hybrid routing: addressed memory for frequent records, retrieval for long-tail evidence and provenance, and shared weights for stable transformations.

Conclusion

User as Engram separates per-user content from shared reasoning by writing facts to hash-addressed rows and learning one adapter. Across experiments, it matches direct recall, improves indirect reasoning $5.6\times$ over per-user LoRA, and uses $161\times$ less state at 100 facts while enabling training-free insertion and deletion. Its token-granular lookup removes external memory hops and remains robust as retrieval recall declines.

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